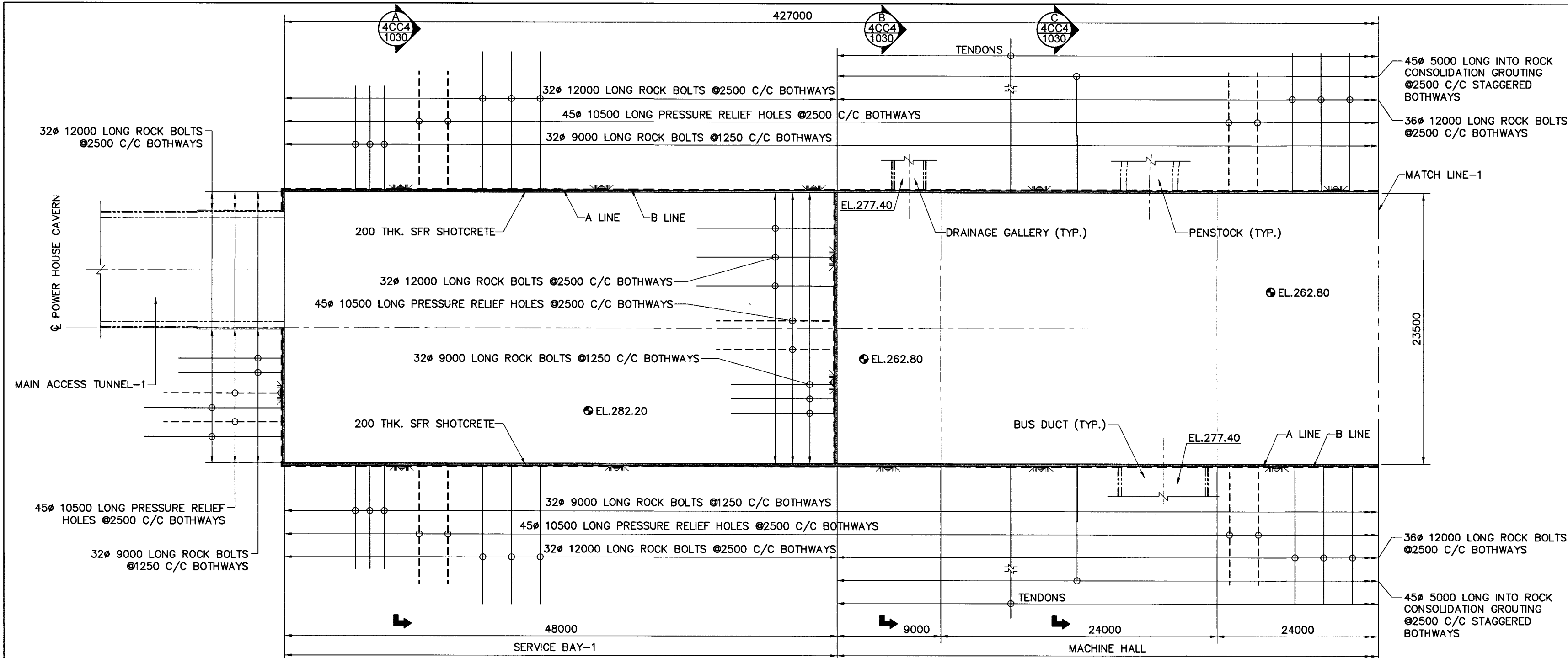
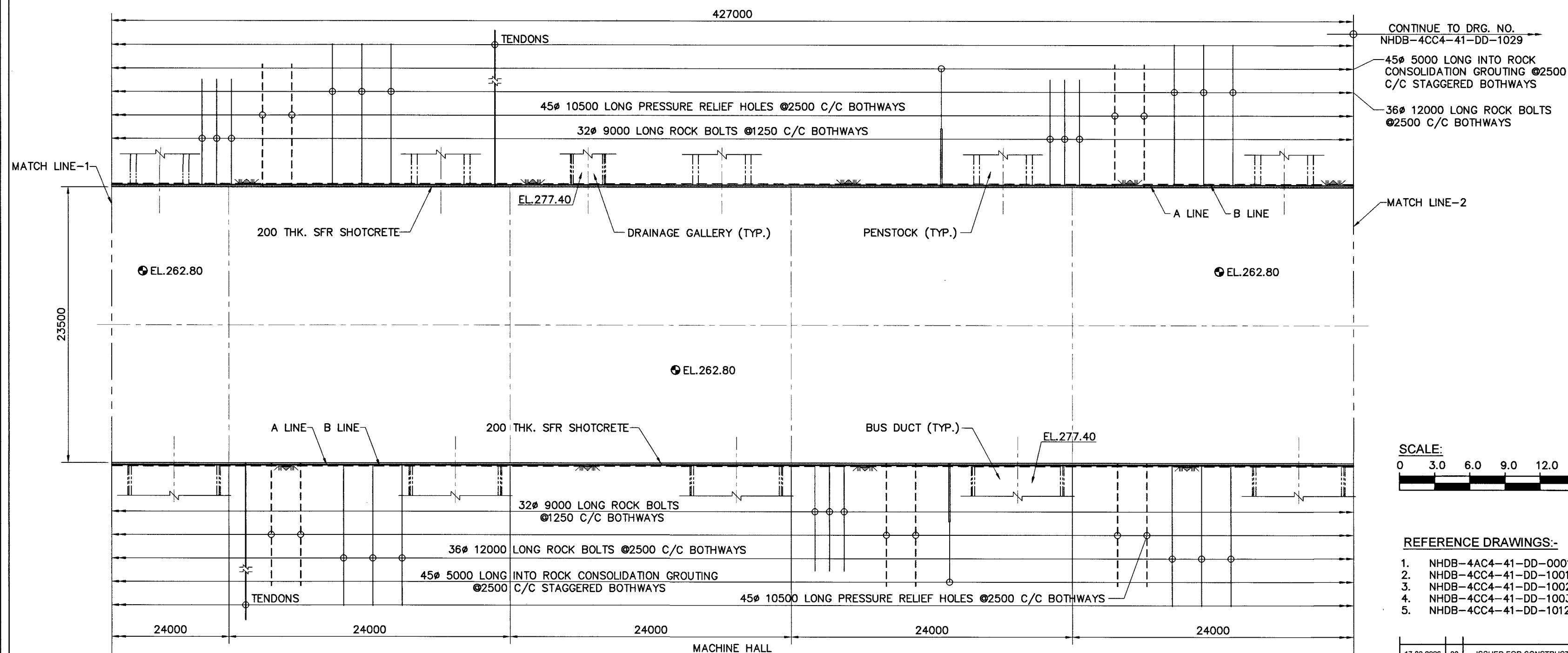




PLAN AT EL. 282.20 / EL. 262.80



PLAN AT EL. 282.20 / EL. 262.80

SCALE:

0 3.0 6.0 9.0 12.0 15.0 METRE

REFERENCE DRAWINGS:-

1. NHDB-4AC4-41-DD-0001.
2. NHDB-4CC4-41-DD-1001.
3. NHDB-4CC4-41-DD-1002.
4. NHDB-4CC4-41-DD-1003 TO 1005.
5. NHDB-4CC4-41-DD-1012 TO 1025.

17.03.2026	00	ISSUED FOR CONSTRUCTION	BY	CH	DEEPAK	SUBMITTED	RECOMMENDED	APPROVED	DEM / HM	अनुमोदित
DATE	NO.	REVISION OR ISSUES	BY	CH	DEEPAK	अनुमोदित	अनुमोदित	अनुमोदित	अनुमोदित	अनुमोदित
DATE	NO.	REVISION OR ISSUES	BY	CH	DEEPAK	अनुमोदित	अनुमोदित	अनुमोदित	अनुमोदित	अनुमोदित
DATE	NO.	REVISION OR ISSUES	BY	CH	DEEPAK	अनुमोदित	अनुमोदित	अनुमोदित	अनुमोदित	अनुमोदित

NOTES:

1. सभी माप मिलीमीटर में तथा ऊचाई मीटर में दिए गए हैं।
2. ALL DIMENSIONS ARE IN MILLIMETRE AND ELEVATIONS ARE IN METRE.
3. THIS DRAWING IS FOR EXCAVATION & ROCK SUPPORT OF POWER HOUSE CAVERN EXCAVATION FROM EL.288.25 TO EL.262.80. FOR EXCAVATION & ROCK SUPPORT BELOW EL.262.80 REFER SEPARATE DRAWING.
4. EXCAVATION SHALL BE DONE IN STAGES CONFORMING TO SECTION B.3.8.7 OF TECHNICAL SPECIFICATIONS. BEFORE TAKING UP THE EXCAVATION OF CAVERN, EXCAVATION METHODOLOGY SHALL BE SUBMITTED BY THE CONTRACTOR AND APPROVED BY THE ENGINEER-IN-CHARGE.
5. EXCAVATION SEQUENCE MAY MODIFY TO SUIT THE SITE CONDITIONS.
6. THE DEFINITION OF 'A' AND 'B' LINE SHALL BE AS PER SECTION B.3 OF TECHNICAL SPECIFICATIONS.
7. NEXT STAGE EXCAVATION SHALL START ONLY AFTER INSTALLATION OF ROCK SUPPORT & INSTRUMENTS IN THE PREVIOUS STAGE INCLUDING GROUTING OF THE ROCK BOLTS.
8. THIS DRAWING SHOWS FINAL ROCK SUPPORT SYSTEM ONLY. IN ADDITION SOME TEMPORARY SUPPORTS MAY BE REQUIRED DURING EXCAVATION BEFORE INSTALLATION OF THE FINAL SUPPORTS. THE SAME MAY BE INSTALLED DEPENDING UPON SITE CONDITIONS WITH THE APPROVAL OF ENGINEER-IN-CHARGE.
9. IN MACHINE HALL & SERVICE BAY-2 (U/S WALL) BETWEEN EL.288.25 TO EL.262.80, 45# 5000 LONG (INTO ROCK) CONSOLIDATION GROUTING @2500 C/C STAGGERED BOTHWAYS SHALL BE PROVIDED THROUGHOUT AS PER SECTION B.7 OF TECHNICAL SPECIFICATIONS. IN SERVICE BAY-1, SERVICE BAY-2 (D/S WALL) & CONTROL BLOCK, 45# 5000 LONG (INTO ROCK) CONSOLIDATION GROUTING @2500 C/C STAGGERED BOTHWAYS SHALL BE PROVIDED IN WEAK ROCK MASS AS DIRECTED BY ENGINEER-IN-CHARGE.
10. THE LOCATION, ORIENTATION AND DEPTH OF THE DRILL HOLES FOR THE CONSOLIDATION GROUTING ARE TENTATIVELY INDICATED ON THE DRAWINGS. THE FINAL NO., LENGTH, LOCATION AND INCLINATION OF THE DRILL HOLES AS WELL AS THE COMPOSITION AND CONSISTENCY OF THE GROUT MIXES, GROUTING PRESSURE, PUMPING RATES AND SEQUENCE IN WHICH THE HOLES ARE TO BE DRILLED AND GROUTED SHALL BE GOVERNED BY ACTUAL CONDITIONS ENCOUNTERED ON SITE AND SHALL BE AT ALL TIMES SUBJECT TO THE APPROVAL BY THE ENGINEER-IN-CHARGE.
11. LINE DRILLING SHALL BE USED ALONG WITH CONTROL BLASTING FOR EXCAVATION TO ACHIEVE THE DESIRED EXCAVATION PROFILE. LINE DRILLING SHALL BE CARRIED OUT CONFORMING TO CLAUSE 3.8.2.2 OF SECTION B.3 OF TECHNICAL SPECIFICATION AS DIRECTED BY ENGINEER-IN-CHARGE.
12. THE BENCHING EXCAVATION SHALL IN NO CASE BE MORE THAN 3000 HEIGHT UNLESS IT IS SUPPORTED BY THE PERMANENT SUPPORT SHOWN IN THE DRAWINGS.
13. THE ROCK BOLTS SHOWN IN THE DRAWING SHALL BE CONFORMING TO SECTION B.4 OF TECHNICAL SPECIFICATIONS. THE ROCK BOLTS SHALL BE TENSIONED AS PER CLAUSE 4.5.8 OF SECTION B.4 OF TECHNICAL SPECIFICATIONS. MINOR MODIFICATION IN THE LENGTH, ORIENTATION, PATTERN AND LOCATION OF ROCK BOLTS MAY BE DONE TO SUIT THE SITE CONDITIONS WITH THE APPROVAL OF ENGINEER-IN-CHARGE. ADDITIONALLY, IN CASE OF PROMINENT SHEAR ZONES, IF ANY, PRE GROUTING SHALL BE PERFORMED AS PER SITE CONDITION IN CONSULTATION WITH SITE GEOLOGIST AS PER SECTION B.4 OF TECHNICAL SPECIFICATION AND AS APPROVED BY ENGINEER-IN-CHARGE.
14. IN CASE OF WEAK ROCK MASS WHERE END ANCHORAGE WITH EXPANSION SHELL ROCK BOLT IS NOT POSSIBLE, RESIN GROUTED ROCK BOLTS SHALL BE PROVIDED IN CONSULTATION WITH SITE GEOLOGIST AS PER DIRECTION OF ENGINEER-IN-CHARGE.
15. THE ROCK BOLTS SHALL BE INSTALLED ONLY ON FINISHED EXCAVATION SURFACES UNLESS AS GOVERNED BY GEOLOGICAL CONDITION AS APPROVED BY ENGINEER-IN-CHARGE.
16. THE CONTRACTOR SHALL EXERCISE ALL PRECAUTIONS TO PREVENT DAMAGE TO ALREADY PLACED SHOTCRETE DURING THE BLASTING FOR THE NEXT PHASE OF EXCAVATION.
17. ATLEAST 50% BOLTS SHALL FORM THE PART OF CYCLE. BALANCE BOLTS CAN BE INSTALLED OUT OF CYCLE BUT BEFORE COMPLETING A PARTICULAR STAGE OF EXCAVATION. THE SPACING BETWEEN THE TWO BOLTS SHALL IN NO CASE EXCEED 2500.
18. INSTALLATION METHODS OF ROCK BOLTS AND SHOTCRETE SHALL BE FULLY OPTIMIZED THROUGH TEST IN THE REPRESENTATIVE ROCK, PRIOR TO ACTUAL START OF WORK.
19. SFR SHOTCRETE SHOWN IN THE DRAWING SHALL BE APPLIED AS SPECIFIED IN SECTION B.5 OF TECHNICAL SPECIFICATION. IF REQUIRED, PLAIN SHOTCRETE WITH WIREMESH CAN BE USED AS PER DIRECTION OF ENGINEER-IN-CHARGE.
20. THE PRESSURE RELIEF HOLES SHOWN IN THE DRAWING SHALL CONFORM TO B.7 OF TECHNICAL SPECIFICATIONS. 36# SLOTTED PVC LINERS IN PRESSURE RELIEF HOLES SHALL BE INSTALLED AS PER THE DIRECTION OF ENGINEER-IN-CHARGE.
21. FOR DETAILS OF PRESSURE RELIEF HOLES ARRANGEMENTS AND INSTRUMENTATION REFER DRAWING NO. NHDB-4CC4-41-DD-1012 TO 1025.
22. POST TENSIONED ROCK BOLTS (TENDONS), THEIR INSTALLATION, TENSIONING AND TESTING SHALL CONFORM TO SECTION B.4 OF TECHNICAL SPECIFICATIONS. FOR TYPE, LOCATION AND LEVELS OF TECHNICAL SPECIFICATIONS. REFER DRAWING NO. NHDB-4CC4-41-DD-1012 TO 1025.

एन एच पी सी लिमिटेड
(भारत सरकार का उद्यम)
NHPC Limited
(A GOVT. OF INDIA ENTERPRISE)

दिबांग बहुउद्देशीय परियोजना
DIBANG MULTIPURPOSE PROJECT

POWER HOUSE CAVERN
EXCAVATION AND ROCK SUPPORTS
FROM EL. 288.25 TO 262.80

SHEET 1 OF 3